

Memory-Centric Agentic Inference — cycles 7-9 clone 2

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Contents

Memory-Centric Agentic Inference — cycles 7-9 clone 2	1
Abstract	1
Introduction	1
Methodology	2
Results	2
Integrated Measurement Design	3
Validation and Governance Status	4
Discussion	5
Conclusions and Recommendations	5
References	5
Appendix: Implementation Details	6

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Abstract

Cycles 7-9 moved clone 2 from a stable branch-level design into validated parent integration for M-EXP-1. The clone-2 artifact, `memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md`, remained unchanged at 308 lines because previous cycles had already converged on the semantic-cache and durable-state risk measurement design. The new work was integration and governance: the parent measurement CSV harness now includes the clone-2 DC-004 semantic-cache rows and DC-003 durable-state rows, the relevant ledger events register the managed artifacts, and final validation is green.

The main outcome is that clone 2 is now validated at three levels: branch artifact, parent CSV ingestion, and parent/root artifact governance. The parent harness preserves the core mechanism boundary: Option B object reuse depends on safe valid semantic retained value, not raw semantic-cache hit rate; Option C durable replay depends on tail-safe durable replay value, not median durable read latency. The final audit for cycle 9 found one governance issue outside clone-2 design: two newly generated clone-0 report artifacts were unregistered. The auditor registered them in ledger event `523f4e76-04d4-4002-8000-000000000205`, after which `promise_check` and `org_check` passed.

Introduction

The root project investigates whether future agentic LLM infrastructure should be organized around memory movement, placement, reuse, compression, and lifetime management rather than arithmetic throughput alone. Clone 2 owns one scoped part of that investigation: risk measurement for semantic caches and durable state.

The branch is tied to two deferred constants:

Deferred constant	Scope	Architecture implication
DC-004	Semantic-cache correctness and invalidation cost	Option B for object reuse is valid only if approximate semantic reuse survives correctness, freshness, provenance, tenant/cache-salt isolation, poisoning, false-positive, and recovery checks.
DC-003	Durable object-store latency distributions for agent state	Option C for durable replay is valid only if workspace, checkpoint, summary-pointer, and trajectory replay paths avoid p50/p95/p99 tail costs that exceed retained value.

Cycles 4-6 established that the branch-level design was stable. Cycles 7-9 answered the next integration question: whether that design had been ingested into the parent M-EXP-1 measurement harness without being flattened into sibling work such as DC-005 trajectory reuse or DC-006 provenance overhead.

Methodology

This report consolidates the supplied cycle records and workspace artifacts. It does not re-audit the results. The primary sources are the researcher, worker, and auditor sessions for cycles 7-9:

Cycle	Researcher	Worker	Auditor
7	9fd22e4a-13f1-4956-9368-8b69db5bcccf	0bf02c53-b17d-4965-ab0f-66929ab69be6	4298f291-56c4-4289-a0ad-f06969257599
8	af2e8d75-4285-45ce-8aa1-7243855567f1	af0a0-4e2f-a380-d63952337341	93-4aed-91fb-b61c63cef
9	f2f1153d-6ec0-45d2-adb3-574607097296	3e-4b2f-9668-9352168167f9	82-4977-91e5-617c4c00b

The current workspace was also inspected for the stable artifact, parent measurement CSVs, references, figures, and validation state. No clone-2-specific figures were produced in cycles 7-9, so no figures are embedded in this report.

Results

Cycle 7: Parent CSV Ingestion

Cycle 7 converted clone 2 from stable design input into parent measurement-harness content.

The researcher session 9fd22e4a-13f1-4956-9368-8b69db5bcccf stated that no new branch artifact work was needed. Instead, it identified the active task as parent M-EXP-1 integration. The sufficiency criteria were concrete: all 11 CDR experiment rows, all 10 CDR thresholds, nonempty instrumentation fields, claim-update links, and downgrade semantics had to appear in the parent CSV harness.

The worker session 0bf02c53-b17d-4965-ab0f-66929ab69be6 patched the parent harness files:

- data/measurement_experiment_specs.csv
- data/measurement_required_fields.csv
- data/measurement_thresholds.csv
- data/measurement_claim_update_matrix.csv
- data/measurement_synthetic_probe_results.csv
- promise_ledger.jsonl

The worker did not edit memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md.

The worker’s coverage probe reported 11 CDR experiment rows, 10 CDR threshold rows, 6 DC-004 rows, 5 DC-003 rows, no empty CDR instrumentation rows, and claim links for CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010.

The auditor session 4298f291-56c4-4289-a0ad-f06969257599 validated the integration. It noted that the parent schema uses `required_instrumentation`, while some session language referred to `instrumentation_fields`; the

auditor treated this as a naming mismatch only, because the populated column carried the required meaning. The same audit confirmed ledger event 523f4e76-04d4-4002-8000-000000000202 as the registered event for the five measurement CSV artifacts.

Cycle 8: Root Artifact Governance Cleanup

Cycle 8 moved from clone-2 ingestion to parent/root hygiene.

The researcher session af2e8d75-4285-45ce-8aa1-71435cbc7fdf stated that clone 2 was complete at both artifact and parent-ingestion levels. The remaining active issue was not design correctness; it was resolving or ledger-documenting orphan managed-path artifacts outside clone-2 scope.

The worker session 2fd25556-71a0-4e2f-a380-dfd5dc23734c registered remaining parent/root artifacts in `promise_ledger.jsonl`:

- 523f4e76-04d4-4002-8000-000000000203: DC-005 merge-readiness verifier and result CSV.
- 523f4e76-04d4-4002-8000-000000000204: fanout cycle reports for clone 0 and clone 2.

The worker also ran `tests/verify_dc005_merge_ready.py`, compiled that verifier, and reran the repository checks. The reported result was `promise_check` green, `org_check` green, and a passing DC-005 verifier.

The auditor session 56395283-f193-4aed-91fb-b61c63cef985 validated the cleanup. It confirmed that the remaining governance warnings had been resolved, that clone-2 rows were still intact, and that DC-003/DC-004 stayed separate from sibling DC-005/DC-006 scope.

Cycle 9: Closure Confirmation and Final Governance Fix

Cycle 9 reconfirmed that clone 2 was fully converged.

The researcher session f2f1153d-6ec0-45d2-adb3-53460fd9d226 stated that clone 2 was validated at branch artifact, parent CSV ingestion, and parent/root artifact-governance levels. It recommended no clone-2 build work, and directed future effort to parent M-EXP-1 synthesis only.

The worker session 17fb7767-6b8e-4b2f-9668-995218167f51 performed no build work and did not edit the clone artifact. It reran `promise_check`, `org_check`, and a targeted CDR preservation probe. The reported preservation state remained 11 CDR experiment rows, 10 thresholds, 6 DC-004 rows, 5 DC-003 rows, and no empty `required_instrumentation` rows.

The auditor session 3b20c8ca-e982-4977-91e5-617c4c00bf10 found one moderate issue, then fixed it. The worker had reported `promise_check` green, but an independent auditor run found two newly generated clone-0 report artifacts that were not ledger-registered:

- `reports/cycles/report_cycles_7-9_clone_0.md`
- `reports/cycles/report_cycles_7-9_clone_0.pdf`

The auditor registered those artifacts in ledger event 523f4e76-04d4-4002-8000-000000000205. After that, `promise_check` reported 67 events, 14 plan milestones, and green status. `org_check` also passed. The same audit reconfirmed that all CDR rows and thresholds were present, semantic terms were intact, durable terms were intact, and no critical or moderate clone-2 issues remained.

Integrated Measurement Design

The stable branch artifact remains `memory-centric-agentic/experiments/cache_durable_risk_measurement_plan.md`. It is a 308-line measurement design for deciding when semantic reuse or durable replay is safe enough to support memory-centric architecture options.

DC-004: Semantic-Cache Risk

The semantic-cache side measures whether approximate reuse is safe after correctness and security checks. The central quantity is safe semantic retained value:

$$V_{\text{semantic safe}} = P_{\text{hit}} \cdot P_{\text{valid|hit}} \cdot C_{\text{recompute avoided}} - C_{\text{lookup}} - C_{\text{validation}} - C_{\text{invalidation}} - C_{\text{recovery}} - E[C_{\text{wrong reuse}}]$$

The parent harness now includes six DC-004 experiment rows:

ID	Measurement purpose
CDR-SEM-001	Raw semantic hit rate versus valid-hit rate.
CDR-SEM-002	Semantic false-positive rate by match-score bucket.
CDR-SEM-003	Stale-hit rate after source-version and invalidation changes.
CDR-SEM-004	Safe hit rate after tenant-scope and cache-salt enforcement.
CDR-SEM-005	Recovery latency and action after invalid cache hit.
CDR-SEM-006	Poisoning or untrusted-provenance rejection rate.

The corresponding thresholds are T-SEM-VALID-001, T-SEM-STALE-002, T-SEM-ISOLATION-003, T-SEM-POISON-004, and T-SEM-RECOVERY-005.

The interpretation preserved through cycles 7-9 is that Option B is not justified by raw hit rate. It requires positive safe value after correctness labels, freshness checks, provenance, tenant/cache-salt isolation, poisoning rejection, validation cost, invalidation cost, recovery cost, and wrong-reuse severity.

DC-003: Durable-State Replay Risk

The durable-state side measures whether remote or persistent replay is fast and valid enough to support Option C. The branch artifact defines durable replay value as:

$$V_{\text{durable replay}} = P_{\text{replay}} \cdot C_{\text{recompute avoided}} - L_{\text{store read}}(p) - L_{\text{consistency}}(p) - C_{\text{pointer validation}} - C_{\text{reconstruction}} - E[C_{\text{replay failure}}]$$

The percentile variable p must include p50, p95, and p99 because long agentic runs can be blocked by the slowest required replay object. Durable replay is therefore not a median-latency claim.

The parent harness now includes five DC-003 experiment rows:

ID	Measurement purpose
CDR-DUR-001	Durable workspace read/write p50, p95, and p99 by size and consistency mode.
CDR-DUR-002	Dependency-path maximum replay latency under fan-in.
CDR-DUR-003	Summary-pointer recovery latency and validity.
CDR-DUR-004	Retention-horizon and hold-state violations.
CDR-DUR-005	Replay success rate under object-store tail injection.

The corresponding thresholds are T-DUR-TAIL-001, T-DUR-P99-002, T-DUR-FANIN-003, T-DUR-RECOVERY-004, and T-DUR-RETENTION-005.

The interpretation preserved through cycles 7-9 is that Option C is not justified by median durable read latency. It requires replay value to survive p95/p99 tails, dependency fan-in, consistency wait, pointer validity, reconstruction latency, retention state, authorization, and replay-failure risk.

Validation and Governance Status

By the end of cycle 9, clone 2 had passed the relevant validation checks at all required levels.

Layer	Status	Evidence
Branch artifact	Validated	<code>cache_durable_risk_measurement_plan.md</code> remained stable at 308 lines.

Layer	Status	Evidence
Parent CSV ingestion	Validated	All 11 CDR experiments, all 10 thresholds, required fields, claim links, and synthetic probes were present.
Root artifact governance	Validated	Ledger events ...000000000202, ...203, ...204, and ...205 registered the relevant managed artifacts.
Repository checks	Validated	Final <code>promise_check</code> reported 67 events, 14 plan milestones, and green status; <code>org_check</code> was green.

The cycle 9 audit explicitly fixed the only newly surfaced moderate issue: unregistered clone-0 cycle 7-9 report artifacts. That issue was outside clone-2 design and outside CDR ingestion, but it affected root governance. After event 523f4e76-04d4-4002-8000-000000000205, no critical or moderate clone-2 issues remained.

Discussion

Cycles 7-9 changed the status of clone 2 from “validated design input” to “validated parent-ingested measurement branch.”

The most important technical preservation is the mechanism boundary. The branch remains about risk-gated reuse:

- Semantic-cache reuse is credited only when valid-hit retained value remains positive after safety and recovery costs.
- Durable-state replay is credited only when replay retained value remains positive after tail latency, fan-in, pointer, consistency, reconstruction, retention, and failure penalties.

This boundary matters because raw reuse signals can overstate memory-centric value. A semantic cache with high hit rate can still be unsafe if hits are stale, poisoned, cross-tenant, or expensive to validate. A durable object store with good median reads can still be unsuitable for Option C if p99 replay tails miss verifier or merge deadlines. The cycles 7-9 integration preserved those distinctions in machine-readable parent CSV rows rather than leaving them only in prose.

The work also clarified ownership. Clone 2 does not own DC-005 trajectory reuse, DC-006 provenance overhead, or full parent synthesis. It supplies stable DC-003 and DC-004 evidence for parent M-EXP-1.

Conclusions and Recommendations

Clone 2 should be treated as closed unless a future parent audit finds a concrete regression in CDR rows, thresholds, fields, or downgrade semantics.

The parent/root synthesis should use registered CDR integration event 523f4e76-04d4-4002-8000-000000000202. It should not cite the earlier absent event 523f4e76-04d4-4002-8000-000000000002.

Future parent synthesis should preserve these rules:

- Option B requires safe valid semantic retained value, not raw semantic hit rate.
- Option C requires tail-safe durable replay value, not median durable read latency.
- DC-003 and DC-004 must remain separate from sibling DC-005 and DC-006 scope.
- Unsafe reuse should downgrade or forbid the architecture option rather than count as a degraded success case.

References

- [6] NVM Express, “Specifications,” NVM Express, accessed 2026-05-11. <https://nvmexpress.org/specifications/>
- [12] vLLM Project, “Automatic Prefix Caching,” vLLM Documentation, accessed 2026-05-11. https://docs.vllm.ai/en/latest/design/prefix_caching/

[13] Sajal Regmi and Chetan Phakami Pun, “GPT Semantic Cache: Reducing LLM Costs and Latency via Semantic Embedding Caching,” arXiv, 2024. <https://arxiv.org/abs/2411.05276>

Appendix: Implementation Details

Code and Artifact Organization

The required clone-2 artifact is:

File	Lines	Purpose
<code>memory-centric-agentic/experiments/cache_durable_risk_measurement_plan</code>	308	Validated measurement design for DC-004 semantic-cache correctness/invalidation risk and DC-003 durable replay-tail risk.

The parent measurement CSVs after cycles 7-9 are:

File	Lines	Cycle 7-9 role
<code>data/measurement_experiment_specs.csv</code>	19	Contains DC-005 rows plus 6 CDR-SEM-* and 5 CDR-DUR-* rows.
<code>data/measurement_required_fields.csv</code>	23	Contains DC-005 required fields plus clone-2 semantic-cache and durable-state gates.
<code>data/measurement_thresholds.csv</code>	15	Contains DC-005 thresholds plus 5 T-SEM-* and 5 T-DUR-* thresholds.
<code>data/measurement_claim_update_matrix.csv</code>	23	Maps clone-2 evidence to CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010.
<code>data/measurement_synthetic_probe_results.csv</code>	13	Contains semantic-cache failure probes and durable replay-tail probes.

The workspace contains 25 Python scripts, 4 Wolfram scripts, and 7,376 total script lines. The broader project contains 29 figures, but none were generated specifically for clone-2 cycles 7-9.

Validation Commands

The reporter pass reran the repository checks after updating MANIFEST.md:

```
python3 -m long_exposure.tools.promise_check <workspace>
python3 -m long_exposure.tools.org_check <workspace>
```

Results:

```
events: 67, plan milestones: 14
OK: promise_check green.
```

```
root files: 5, root dirs: 10; standard folders present: ['audits', 'data', 'docs', 'reports', 'scripts', 's']
OK: org_check green.
```

A `git diff` command was attempted for manifest review, but `<workspace>` is not a Git repository. This did not affect validation; both project checks passed after the manifest update.

Manifest Update

MANIFEST.md was updated as a workspace snapshot. The update changed the snapshot description from the prior DC-005 terminal handoff state to the clone-2 cycles 7-9 DC-003/DC-004 parent-ingestion and validation state. It also updated the line counts and purposes for the five parent measurement CSVs, and added cross-references from `cache_durable_risk_measurement_plan.md` into the parent experiment, threshold, required-field, and claim-update CSVs.

No **## Key Files** preservation section was present in the manifest, so no protected section needed to be preserved.

Session Reference Map

Session	Role in report
9fd22e4a-13f1-4956-9368-8b69db5bcccf	Cycle 7 researcher brief defining parent CDR integration sufficiency criteria.
0bf02c53-b17d-4965-ab0f-66929ab69be6	Cycle 7 worker output for parent CSV ingestion and CDR coverage probe.
4298f291-56c4-4289-a0ad-f06969257599	Cycle 7 audit validating parent CSV ingestion and noting remaining non-clone orphan warnings.
af2e8d75-4285-45ce-8aa1-71435cbc7fdf	Cycle 8 researcher brief shifting active work to root artifact governance.
2fd25556-71a0-4e2f-a380-dfd5dc23734c	Cycle 8 worker output registering DC-005 verifier artifacts and fanout report artifacts.
56395283-f193-4aed-91fb-b61c63cef985	Cycle 8 audit validating root governance cleanup and clone-2 preservation.
f2f1153d-6ec0-45d2-adb3-53460fd9d226	Cycle 9 researcher brief declaring clone-2 fully converged.
17fb7767-6b8e-4b2f-9668-995218167f51	Cycle 9 worker output rerunning green checks and CDR preservation probe.
3b20c8ca-e982-4977-91e5-617c4c00bf10	Cycle 9 audit fixing clone-0 report artifact registration and issuing final validation.

Cross-Reference Map

Origin	Consuming artifact	Flow
memory-centric-agentic/experiments	<code>data/measurement/experiment_exports.csv</code>	CDR-SEM-* and CDR-DUR-* measurement designs become parent M-EXP-1 experiment rows.
memory-centric-agentic/experiments	<code>data/measurement/required_fields.jsonl</code>	Schema and cache and durable-state instrumentation gates become required parent fields.
memory-centric-agentic/experiments	<code>data/measurement/thresholds.jsonl</code>	T-SEM-* and T-DUR-* downgrade/forbid rules become parent thresholds.
memory-centric-agentic/experiments	<code>data/measurement/claim_updates.jsonl</code>	Parent evidence updates claims CL-002, CL-003, CL-006, CL-008, CL-009, and CL-010.
memory-centric-agentic/experiments	<code>data/measurement/synthetic_mechanism_checks.jsonl</code>	Schema and cache false-positive/stale/recovery probes and durable tail/pointer/retention probes become synthetic mechanism checks.
promise_ledger.jsonl event 523f4e76-04d4-4002-8000-000000000202	Parent M-EXP-1 governance	Registers the five CDR measurement CSV artifacts.

Origin	Consuming artifact	Flow
promise_ledger.jsonl event 523f4e76-04d4-4002-8000-000000000205	Final cycle 9 validation	Registers newly surfaced clone-0 cycle 7-9 report artifacts, restoring green promise_check .